Date: 31 Jul 2026 Time: 15:38 51s

No.	Section Description.	Questions	Max	Mark
1	This part will provide an overview of the entire project by outlining the project's history as well as the suggested Vision, Goals, and Timelines. Essentially, it establishes the project's context and makes the paper compelling enough to read and support, allowing the project to get underway.	7	50	

The project will be considered successful when all of the following criteria are met:

1. Platform delivered on time - The web-based reporting system is fully built, integrated, and ready for demonstration by November 2026, within the ISJ107V academic project schedule.

2. Core functionality working - Users can register an account, log in securely, submit a missing children report with all required fields and an optional photograph, and view or search existing active reports without errors.

3. Security and data protection in place - All user passwords are stored as BCrypt hashes. Every API endpoint requires valid JWT authentication. No personally identifiable information about missing children is accessible to unauthenticated users under any circumstances.

4. Alert notifications and search working - The system successfully sends geographic alert notifications to registered users within 1 minute of a new report being submitted. The search function returns accurate results within 2 seconds.

5. AI duplicate detection and analytics working - The AI-based duplicate detection module achieves a precision of at least 80% on a prepared test dataset. The analytics dashboard correctly displays reporting trends by location, time period, and resolution status.

1	1. The success of the project is the goal of every project manager. However, what exactly constitutes success? Is it the timely and cost-effective completion of the project or the production of project deliverables at the anticipated level of quality? Is the client satisfied with the outcome and was kept updated throughout the project? Or is it just that the opportunity has materialized as anticipated or that the business issue has been resolved? The success criterion must be carefully considered before being specified here. We recommend outlining five essential success criteria that will be used to gauge the project's progress.	10	0

Missing children's cases have always been a serious problem in South Africa, but the way communities respond to them has not really changed much over the years. Missing Children South Africa, a non-profit organisation that has been tracking these cases since the early 2000s, consistently records around 1 697 missing children per year (Missing Children South Africa, 2024). That number has remained concerning, and the response systems available to ordinary community members have not kept up.

Before social media, communities relied on radio announcements, newspaper notices, and word of mouth through churches and community centres. When platforms like Facebook and WhatsApp became popular in the 2010s, things improved in some ways – information could reach more people faster. But it also created new problems. The same case would get posted to dozens of groups by different people, sometimes with different photos or slightly different details, and there was no way to update all of them when the child was eventually found.

The official route – reporting to the South African Police Service (SAPS) – is still the most formal option, but police resources are limited and information reported at a police station does not automatically reach the community. There is a gap between the official system and what communities can actually access and use.

Looking at other countries, there are better examples. The United States has the National Center for

Missing and Exploited Children (NCMEC) which runs a centralised online database and alert system. Many European countries use systems like the Amber Alert. South Africa has nothing like this designed specifically for community-level reporting and response. That gap is what this project is trying to address.

2	2.1 To set the stage for the project, you should provide the reader with a quick overview of the business's background before delving deeply into the proposal. Since this serves as the foundation for the project, you must outline all pertinent historical events up to this point.	5	2

The main problem is that there is no centralised, structured digital platform for South African communities to report and track missing children. This creates several interconnected issues that slow down the community's ability to respond effectively.

1. Information gets scattered across too many platforms - When a child goes missing, people start posting on WhatsApp groups, Facebook pages, Twitter, and community notice boards all at the same time. None of these platforms talk to each other. Someone who is only on one platform might miss the report entirely, and someone who is on five platforms ends up seeing five slightly different versions of the same information.

2. The same case gets reported multiple times - Because there is no single place to report, different people often submit the same case to different platforms without knowing someone else already did. Over time this creates multiple conflicting versions of the same report — different photos, different descriptions, different locations — which makes it difficult for anyone trying to help to know which version is accurate.

3. No way to identify patterns or high-risk areas - Because the data is spread across so many unconnected sources, it is impossible to analyse whether certain areas have more missing children's cases, or whether cases tend to happen more at certain times of year. Communities and organisations cannot use scattered data to plan or identify where they should be paying more attention.

4. No structured way to alert the right people - When a new case is reported, there is currently no automatic way to notify community members who live near where the child was last seen. Reaching the right people requires manual posting across multiple platforms, and even then, you can never be sure the information reached everyone it should have.

All these problems together mean that communities respond more slowly and less effectively than they could with a better system in place.

3	2.2.1 Provide a detailed description of the business problem if you have one that this project must address. Describe the nature of the issue and the symptoms, for instance, low client satisfaction, poor communication, or financial difficulties.	5	5

The opportunity here is to build a single, centralised web-based platform that brings all of this together — one place to report, one place to search, and one system to send alerts — within the ISJ107V academic project schedule ending with a demonstration in November 2026.

Why this opportunity exists now - A few years ago, building something like this with AI included would have required a lot of specialist knowledge and expensive software. That has changed. Python libraries like scikit-learn and FastAPI are freely available, well-documented, and powerful enough to build a working AI duplicate detection service within the scope of a student project. The core web development stack — Java Spring Boot, PostgreSQL, HTML/CSS/JavaScript — is well-established and widely supported with learning resources. South Africa also has no equivalent of the international Amber Alert or NCMEC systems at community level, which means there is a clear gap in the market for this type of platform.

The deadline for seizing this opportunity - Based on the statistics from Missing Children South Africa (2024), approximately 140 children are reported missing every month. Every month that passes withou

t a better system means those cases are being handled through fragmented, unreliable channels. The ac
ademic deadline of November 2026 gives a fixed, meaningful target to work towards. Missing that deadl
ine would mean both failing the module requirement and delaying something that could genuinely help c
ommunities.

How it will be achieved - I plan to build the system using Java Spring Boot for the backend, PostgreS
QL for the database, and HTML/CSS/JavaScript for the frontend. The AI duplicate detection module will
 be built separately in Python using FastAPI, and it will communicate with the Java backend through a
 REST API. Development will follow the SDLC methodology across six phases, from planning through to t
he final demonstration in November 2026.

4	2.2.2 Provide a detailed description of the project if it has a business opportunity that needs to be fulfilled. Specify the circumstances around the opportunity, the deadline for seizing it, and the method by which it must be achieved. Opportunities can include things like a gap in the market for a certain good or service, a new technology that can help the company, or a change in regulations that allows the company to offer a new good or service that it was previously unable to.	5	4

Goal of the solution - To develop and deliver a functional AI-enhanced web-based platform for reporti
ng and tracking missing children, ready for demonstration by November 2026.

In scope:

Web-based user registration and login
Missing children report submission with optional image upload
Viewing and searching active reports
Community alert notifications based on geographic area
AI-based duplicate report detection using NLP text similarity analysis
Analytics dashboard displaying reporting trends by location and time period
Secure storage of all user and report data

Out of scope:

Integration with SAPS or police databases
Facial recognition technology
GPS tracking of children
Mobile application development

Approach - The project will follow the Software Development Life Cycle (SDLC) across six phases:

Planning and Requirements Gathering (March 2026) - Define what the system needs to do, identify user
roles, and document requirements.
System Design (April 2026) - Design system architecture, database schema, and UI wireframes.
Backend Development (May 2026) - Build the Spring Boot REST API, implement authentication, and develo
p the Python AI microservice.
Frontend Development and Integration (June 2026) - Build the HTML/CSS/JavaScript interface and connec
t it to the backend.
Testing and Evaluation (July-August 2026) - Test all modules, fix bugs, and evaluate AI accuracy.
Documentation and Final Presentation (September-November 2026) - Write up documentation and prepare t
he live demonstration.

5	2.3 The next step is to outline the opportunity or solution that the initiative must provide. You must clarify the following while outlining the solution: • The solution's goal; • Its scope, or what falls inside or outside of it; • The method by which the answer must be provided (i.e., approach); The project's sole goal is to "deliver the solution to that problem or opportunity within the	10	8

timeframes required." Keep that in mind. The project team will therefore find it easier to understand what needs to be produced and by when if you can provide a detailed description of the solution.

```
Vision Statement - By developing a centralized community-based digital platform for reporting missing
 children, this project aims to improve community awareness and response time by at least 30% within
the development period, enabling faster sharing of information and improving efforts to locate missin
g children in South Africa.

Goal 1: Core Platform Delivery - Design and develop a functional web-based reporting system by Month
4 (June 2026) that allows registered users to submit missing children reports with the child's name,
age, gender, physical description, last known location, date last seen, and an optional photograph.

Specific - A web-based reporting platform with clearly defined input fields and optional image upload
.
Measurable - The system must successfully accept, store, and display at least one complete report end
 to end without errors.
Action-oriented - I will design, develop, and integrate the reporting module including the frontend f
orm, Spring Boot controller, and PostgreSQL database layer.
Realistic - Achievable using Java Spring Boot and PostgreSQL within the available timeline.
Time-based - Backend complete by Month 3 (May 2026), frontend integration complete by Month 4 (June 2
026).

Goal 2: Security and Data Protection - Implement secure authentication and data storage so that only
registered, authenticated users can access or submit reports, with backend security completed by Mont
h 3 (May 2026).

Specific - User registration, login, JWT-based sessions, role-based access control, and BCrypt passwo
rd hashing.
Measurable - No unauthenticated request to a protected endpoint should succeed. All passwords stored
as BCrypt hashes.
Action-oriented - I will implement Spring Security, JWT token validation, and database access control
s.
Realistic - Standard security features fully supported within the Spring Boot framework.
Time-based - All security features complete and tested by end of Month 3 (May 2026).

Goal 3: AI and Analytics Features - Implement an AI-based duplicate detection module that correctly i
dentifies at least 80% of duplicate reports and delivers an analytics dashboard showing reporting tre
nds, both working by Month 4 (June 2026).

Specific - A Python microservice using TF-IDF and cosine similarity, plus a dashboard with at least t
hree analytics views.
Measurable - Duplicate detection precision of at least 80% and recall of at least 75%, tested against
 20 known report pairs.
Action-oriented - I will develop the Python FastAPI microservice, integrate it with Spring Boot, and
build the analytics views.
Realistic - Python libraries scikit-learn and FastAPI are freely available and well-documented.
Time-based - AI module and dashboard integrated and testable by end of Month 4 (June 2026).
```

6	3.1 Describe the project's vision and objectives. They must be "smart," which stands for Specific, Measurable, Action orientated, Realistic and Time based. Make every effort to incorporate the business opportunity or problem into the vision so that the project team is aware of the precise problem the vision is trying to solve. An illustration of a project vision is as follows: “By providing additional services to our valued customers, we hope to increase customer satisfaction by 25% by December 1, 2025”. Then, list 3 goals which relate to your vision above. Again, make them SMART	10	10

The project will be considered successful when all of the following criteria are met:

1. Platform delivered on time - The web-based reporting system is fully built, integrated, and ready for demonstration by November 2026, within the ISJ107V academic project schedule.

2. Core functionality working - Users can register an account, log in securely, submit a missing children report with all required fields and an optional photograph, and view or search existing active reports without errors.

3. Security and data protection in place - All user passwords are stored as BCrypt hashes. Every API endpoint requires valid JWT authentication. No personally identifiable information about missing children is accessible to unauthenticated users under any circumstances.

4. Alert notifications and search working - The system sends geographic alert notifications to registered users within 1 minute of a new report being submitted. The search function returns accurate results within 2 seconds.

5. AI duplicate detection and analytics working - The AI-based duplicate detection module achieves a precision of at least 80% on a prepared test dataset. The analytics dashboard correctly displays reporting trends by location, time period, and resolution status.

7 3.5 The success of the project is the goal of every project manager. However, what exactly constitutes success? 5 5

Is it the timely and cost-effective completion of the project or the production of project deliverables at the anticipated level of quality? Is the client satisfied with the outcome and was kept updated throughout the project? Or is it just that the opportunity has materialized as anticipated or that the business issue has been resolved?

The success criterion must be carefully considered before being specified here. We recommend outlining five essential success criteria that will be used to gauge the project's progress.

No.	Section Description.	Questions	Max	Mark
x	Question 3.2, 3.3 and 3.4	3	20	

3.2 Project Deliverables

Title	Description	Notes
System Requirements Document	Documentation describing functional and non-functional requirements, UML diagrams, database schema, and API design	Produced during planning and design phases
Web-Based Reporting Platform	Functional system for submitting, viewing, and searching for missing children reports	Core system component, completed by June 2026
Database System	Secure PostgreSQL database storing user accounts, reports, duplicate flags, and notifications	Implemented during backend development, May 2026
AI-Based Duplicate Detection Module	Python microservice using TF-IDF and cosine similarity to flag duplicate reports with similarity threshold of 0.75 or above	Integrated with Spring Boot backend by June 2026
Analytics Dashboard	Dashboard displaying report statistics by location, time period, and resolution status	Completed during frontend integration, June 2026
Alert Notification Feature	Automated geographic alerts sent to registered users when a new missing children report is submitted	Implemented during backend development, May 2026
Final Project Documentation	Complete documentation including system design,	Submitted at project completion, October 2026

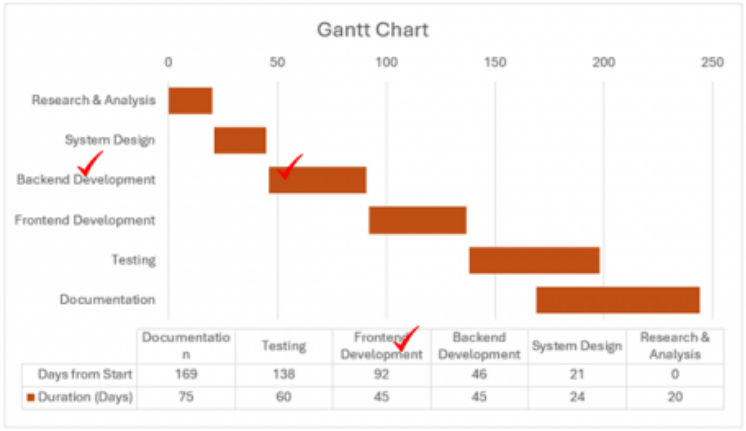
	source code, test results, user manual, and project evaluation	

1 3.2 Deliverables from each project contribute to the solution that addresses the business opportunity or challenge. These deliverables specify the scope of what needs to be produced and are essential to the project's success. As a result, it is imperative that you provide a detailed list of all the important deliverables that must be created. Use a table format below. (submit as an image) 1 10

Task Duration Breakdown:

Task	Start Date	Duration (Days)
Research & Analysis	16-Mar-26	20
System Design	06-Apr-26	24
Backend Development	01-May-26	45
Frontend Development	16-Jun-26	45
Testing	01-Aug-26	60
Documentation	01-Sept-26	75

Gantt Chart



2 3.3 Every project has to be finished by a certain date. To optimize the project's value to the business, every opportunity or business challenge must be resolved within a given timeline. This timeline is essential to the project's success since the firm might not gain the desired degree of benefit if the solution is delivered after the allotted period. Therefore, you must outline the project's delivery timeline. Explain the timeline and the significance of the project's delivering the solution within this time limit. Use Example on image 1.2 (Submit as an image) 1 4

Type	Resource	Quantity	Notes
People	Ntwanano Mathebula (Developer)	1	Full system design, development, testing, and documentation
People	Dr HD Masethe (Lecturer/Supervisor)	1	Academic guidance, feedback on deliverables, and assessment
Tools	IntelliJ IDEA	1	Backend development environment for Java Spring Boot
Tools	Visual Studio Code	1	Frontend development environment for HTML/CSS/JavaScript

Tools	PostgreSQL 15	1	Local database server for development and testing
Tools	Git + GitHub	1	Version control and source code management
Tools	Microsoft Teams	1	Weekly class sessions and communication with supervisor
Supplies	Development laptop/computer	1	Required for all system development and testing
Supplies	Python 3.11 + pip	1	AI module development environment
Supplies	Spring Boot, Spring Security, JPA/Hibernate	1	Backend framework, authentication, and ORM libraries
Supplies	scikit-learn, rapidfuzz, FastAPI, Uvicorn	1	Python AI/ML libraries and microservice
Supplies	TUT EC / Submission Portal	1	Assignment submissions and academic materials

3 3.4 The number of resources needed to complete the project will be a factor in our decision to approve the proposal. Fill in the following table, which lists the general resources people, tools, and supplies necessary to complete the job. The project can use these general numbers to get a sense of the number of resources needed to complete the project. Submit a picture. 1 5

- = end = -

Total Marks [53 / 70]